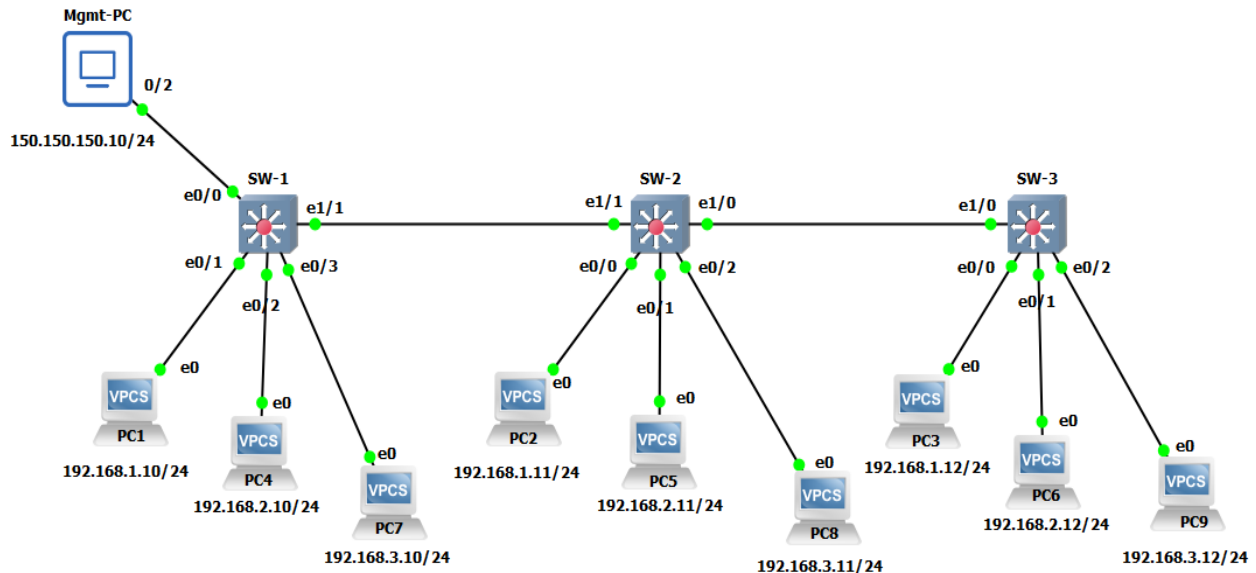


VLAN Trunking Protocol (VTP)



Task

1. Configure the SW-1 with IP 150.150.150.1, SW-2 with IP 150.150.150.2 and SW-3 with IP 150.150.150.3 for device management and configure enable password as **ccna** and device access password as **cisco**.
2. Configure IP address on VPCs
3. Configure trunk links between the switches
4. Configure VTP version 3 with domain name rsatechforum.in on all switches. SW-1 should be VTP server and SW-2 & SW-3 as VTP client.
5. Create VLAN-10 as Sale, VLAN-20 as Production and VLAN-30 as Marketing.
6. Assign switchports in respective VLAN as per the topology
7. Test the connectivity between VPCs in same & different VLAN.
8. Configure SW-2 as VTP Transparent mode.

Note: Mgmt-PC is a cloud which bridge the switch SW-1 with loopback adapter on real PC.

Task-1: Configure the switches with IP address for device management and configure enable password as **ccna** and device access password as **cisco**.

Switch-1 configuration:

```
SW-1#config t
SW-1(config)#hostname SW-1
SW-1(config)#no ip domain-lookup
SW-1(config)#enable password ccna
SW-1(config)#int vlan1
SW-1(config-if)#ip address 150.150.150.1 255.255.255.0
```

```

SW-1(config-if)#no shut
SW-1(config-if)#exit
SW-1(config)#banner motd $
Enter TEXT message. End with the character '$'.
#####
#####
#####          Welcome to RSATechforum          #####
#####          This is Switch-1                  #####
#####
#####
$
SW-1(config)#line vty 0 4
SW-1(config-line)#password cisco
SW-1(config-line)#login
SW-1(config-line)#logging synchronous
SW-1(config-line)#exec-timeout 1 0
SW-1(config-line)#exit
SW-1(config)#line console 0
SW-1(config-line)#password cisco1
SW-1(config-line)#login
SW-1(config-line)#logging synchronous
SW-1(config-line)#exec-timeout 0 0
SW-1(config-line)#exit
SW-1(config)#exit
SW-1#

```

Switch-2 configuration:

```

SW-2#config t
SW-2(config)#hostname SW-2
SW-2(config)#no ip domain-lookup
SW-2(config)#enable password ccna
SW-2(config)#int vlan1
SW-2(config-if)#ip address 150.150.150.2 255.255.255.0
SW-2(config-if)#no shut
SW-2(config-if)#exit
SW-2(config)#banner motd $
Enter TEXT message. End with the character '$'.
#####
#####
#####          Welcome to RSATechforum          #####
#####          This is Switch-2                  #####
#####
#####
$
SW-2(config)#line vty 0 4
SW-2(config-line)#password cisco
SW-2(config-line)#login
SW-2(config-line)#logging synchronous
SW-2(config-line)#exec-timeout 1 0
SW-2(config-line)#exit

```

```
SW-2(config)#line console 0
SW-2(config-line)#password cisco1
SW-2(config-line)#login
SW-2(config-line)#logging synchronous
SW-2(config-line)#exec-timeout 0 0
SW-2(config-line)#exit
SW-2(config)#exit
SW-2#
```

Switch-3 configuration:

```
SW-3#config t
SW-3(config)#hostname SW-3
SW-3(config)#no ip domain-lookup
SW-3(config)#enable password ccna
SW-3(config)#int vlan1
SW-3(config-if)#ip address 150.150.150.3 255.255.255.0
SW-3(config-if)#no shut
SW-3(config-if)#exit
SW-3(config)#banner motd $
Enter TEXT message. End with the character '$'.
#####
#####
#####      Welcome to RSATechforum      #####
#####      This is Switch-3              #####
#####
#####
$
SW-3(config)#line vty 0 4
SW-3(config-line)#password cisco
SW-3(config-line)#login
SW-3(config-line)#logging synchronous
SW-3(config-line)#exec-timeout 1 0
SW-3(config-line)#exit
SW-3(config)#line console 0
SW-3(config-line)#password cisco1
SW-3(config-line)#login
SW-3(config-line)#logging synchronous
SW-3(config-line)#exec-timeout 0 0
SW-3(config-line)#exit
SW-3(config)#exit
SW-3#
```

Task-2: Configure IP address on VPCs

PC-1 Configuration:

```
PC1> ip 192.168.1.10/24 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.10 255.255.255.0 gateway 192.168.1.1
```

PC-2 Configuration:

```
PC2> ip 192.168.1.11/24 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.11 255.255.255.0 gateway 192.168.1.1
```

PC-3 Configuration:

```
PC3> ip 192.168.1.12/24 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.12 255.255.255.0 gateway 192.168.1.1
```

PC-4 Configuration:

```
PC4> ip 192.168.2.10/24 192.168.2.1
Checking for duplicate address...
PC1 : 192.168.2.10 255.255.255.0 gateway 192.168.2.1
```

PC-5 Configuration:

```
PC5> ip 192.168.2.11/24 192.168.2.1
Checking for duplicate address...
PC1 : 192.168.2.11 255.255.255.0 gateway 192.168.2.1
```

PC-6 Configuration:

```
PC6> ip 192.168.2.12/24 192.168.2.1
Checking for duplicate address...
PC1 : 192.168.2.12 255.255.255.0 gateway 192.168.2.1
```

PC-7 Configuration:

```
PC7> ip 192.168.3.10/24 192.168.3.1
Checking for duplicate address...
PC1 : 192.168.3.10 255.255.255.0 gateway 192.168.3.1
```

PC-8 Configuration:

```
PC8> ip 192.168.3.11/24 192.168.3.1
Checking for duplicate address...
PC1 : 192.168.3.11 255.255.255.0 gateway 192.168.3.1
```

PC-9 Configuration:

```
PC9> ip 192.168.3.12/24 192.168.3.1
Checking for duplicate address...
PC1 : 192.168.3.12 255.255.255.0 gateway 192.168.3.1
```

Task-3: Configure trunk links between the switches**Switch-1 configuration:**

```
SW-1#config t
SW-1(config)#int e1/1
SW-1(config-if)#switchport trunk encapsulation dot1q
SW-1(config-if)#switchport mode trunk
SW-1(config-if)#exit
SW-1(config)#exit
SW-1#
```

Switch-2 configuration:

```
SW-2#config t
SW-2(config)#int e1/0
SW-2(config-if)#switchport trunk encapsulation dot1q
SW-2(config-if)#switchport mode trunk
SW-2(config-if)#exit
SW-2(config)#exit
SW-2#
```

✓ Verification & Testing:**SW-1#sh int trunk**

Port	Mode	Encapsulation	Status	Native vlan
Et1/1	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Et1/1	1-4094

Port	Vlans allowed and active in management domain
Et1/1	1

Port	Vlans in spanning tree forwarding state and not pruned
Et1/1	1

SW-2#sh int trunk

Port	Mode	Encapsulation	Status	Native vlan
Et1/0	on	802.1q	trunking	1
Et1/1	auto	n-802.1q	trunking	1

Port	Vlans allowed on trunk
Et1/0	1-4094
Et1/1	1-4094

Port	Vlans allowed and active in management domain
Et1/0	1
Et1/1	1

Port	Vlans in spanning tree forwarding state and not pruned
Et1/0	1
Et1/1	1

Et1/0 1
Et1/1 1

SW-3#sh int trunk

Port	Mode	Encapsulation	Status	Native vlan
Et1/0	auto	n-802.1q	trunking	1

Port	Vlans allowed on trunk
Et1/0	1-4094

Port	Vlans allowed and active in management domain
Et1/0	1

Port	Vlans in spanning tree forwarding state and not pruned
Et1/0	1

Task-4: Configure VTP version 3 with domain name rsatechforum.in on all switches. SW-1 should be VTP server and SW-2 & SW-3 as VTP client.

Server	The Default mode of a switch operation, responsible for creating, modifying and deleting VLANs in your Switched network, residing in the VTP Domain. The VTP Server advertises its VLAN configuration to other switches in the domain.
Client	VTP clients behave the same way as VTP Servers, but cannot create, change or delete VLANs. It just receives VLAN information from the VTP Server Switch in the domain
Transparent	The transparent mode is a way to switches do not participate in VTP. Neither receive nor advertises its VLAN configuration, just forward the VTP packets that other switches (Server and Clients) are exchanging.

Parameter	VTP v1	VTP v2	VTP v3
Mode	Server / Client / Transparent	Server / Client / Transparent	Server / Client / Transparent / Off
Token Ring support	Not Supported	Supported	Supported
Unrecognized Type-Length-Value (TLV) Support	Not Supported	Supported	Supported
Version-Dependent Transparent Mode	In VTP version 1, a VTP transparent network device inspects VTP messages for the domain name and version and forwards a message only if the version and domain name match	VTP version 2 forwards VTP messages in transparent mode without checking the version	VTP version 2 forwards VTP messages in transparent mode without checking the version
Consistency Checks	Not Supported	Supported	Supported
Propagation of extended Vlan range	1 to 1000	1 to 1000	1 to 4094
Private VLANs	Not Supported	Not Supported	Supported
Support for propagation of any database in a domain	a VTP server is used to back up	a VTP server is used to back up	Supported
Authentication	Plain text password	Plain text password	Encrypted password

Switch-1 configuration:

```
SW-1#config t
SW-1(config)#vtp mode server
Device mode already VTP Server for VLANs.
SW-1(config)#vtp domain rsatechforum.in
Changing VTP domain name from NULL to rsatechforum.in
SW-1(config)#vtp version 3
SW-1(config)#vtp password Pass@123
Setting device VTP password to Pass@123
SW-1(config)#exit
SW-1#
```

SW-1#vtp primary

This system is becoming primary server for feature vlan
No conflicting VTP3 devices found.

Do you want to continue? [confirm]

SW-1#

*Dec 6 15:32:45.952: %SW_VLAN-4-VTP_PRIMARY_SERVER_CHG: aabb.cc80.0100 has become the primary server for the VLAN VTP feature

Note: In VTPv3, we need use above command on Server switch to enable VLAN configuration features. Without this command, we cannot create VLANs

Switch-2 configuration:

```
SW-2#config t
SW-2(config)#vtp mode client
Setting device to VTP Client mode for VLANs.
SW-2(config)#vtp domain rsatechforum.in
Changing VTP domain name from NULL to rsatechforum.in
SW-2(config)#vtp version 3
SW-2(config)#vtp password Pass@123
Setting device VTP password to Pass@123
SW-2(config)#exit
SW-2#
```

Switch-3 configuration:

```
SW-3#config t
SW-3(config)#vtp mode client
Setting device to VTP Client mode for VLANs.
SW-3(config)#vtp domain rsatechforum.in
Changing VTP domain name from NULL to rsatechforum.in
SW-3(config)#vtp version 3
SW-3(config)#vtp password Pass@123
Setting device VTP password to Pass@123
SW-3(config)#exit
SW-3(config)#exit
SW-3#
```

✓ Verification & Testing:

SW-1#sh vlan brief

VLAN Name	Status	Ports
1 default	active	Et0/0, Et0/1, Et0/2, Et0/3 Et1/0, Et1/2, Et1/3, Et2/0 Et2/1, Et2/2, Et2/3, Et3/0 Et3/1, Et3/2, Et3/3
1002 fddi-default	act/unsup	
1003 trcrf-default	act/unsup	
1004 fddinet-default	act/unsup	
1005 trbrf-default	act/unsup	
SW-1#		

SW-2#sh vlan brief

VLAN Name	Status	Ports
1 default	active	Et0/0, Et0/1, Et0/2, Et0/3 Et1/2, Et1/3, Et2/0, Et2/1 Et2/2, Et2/3, Et3/0, Et3/1 Et3/2, Et3/3
1002 fddi-default	act/unsup	
1003 trcrf-default	act/unsup	
1004 fddinet-default	act/unsup	
1005 trbrf-default	act/unsup	
SW-2#		

SW-3#sh vlan brief

VLAN Name	Status	Ports
1 default	active	Et0/0, Et0/1, Et0/2, Et0/3 Et1/1, Et1/2, Et1/3, Et2/0 Et2/1, Et2/2, Et2/3, Et3/0 Et3/1, Et3/2, Et3/3
1002 fddi-default	act/unsup	
1003 trcrf-default	act/unsup	
1004 fddinet-default	act/unsup	
1005 trbrf-default	act/unsup	
SW-3#		

SW-1#sh vtp status

VTP Version capable : 1 to 3
VTP version running : 3
 VTP Domain Name : rsatechforum.in
 VTP Pruning Mode : Disabled
 VTP Traps Generation : Disabled
 Device ID : aabb.cc80.0100

Feature VLAN:

VTP Operating Mode : Primary Server
Number of existing VLANs : 5
Number of existing extended VLANs : 0
Maximum VLANs supported locally : 4096
Configuration Revision : 1
Primary ID : aabb.cc80.0100
Primary Description : SW-1
MD5 digest : 0x9A 0xCD 0x85 0xFC 0xB0 0xAA 0x1B 0x2D
0xB0 0x93 0x8E 0x49 0xD2 0x3B 0x63 0x8A

Feature MST:

VTP Operating Mode : Transparent

Feature UNKNOWN:

VTP Operating Mode : Transparent

SW-2#sh vtp status

VTP Version capable : 1 to 3
VTP version running : 3
VTP Domain Name : rsatechforum.in
VTP Pruning Mode : Disabled
VTP Traps Generation : Disabled
Device ID : aabb.cc80.0200

Feature VLAN:

VTP Operating Mode : Client
Number of existing VLANs : 5
Number of existing extended VLANs : 0
Maximum VLANs supported locally : 4096
Configuration Revision : 1
Primary ID : aabb.cc80.0100
Primary Description : SW-1
MD5 digest : 0x9A 0xCD 0x85 0xFC 0xB0 0xAA 0x1B 0x2D
0xB0 0x93 0x8E 0x49 0xD2 0x3B 0x63 0x8A

Feature MST:

VTP Operating Mode : Transparent

Feature UNKNOWN:

VTP Operating Mode : Transparent

SW-3#sh vtp status

VTP Version capable : 1 to 3
VTP version running : 3
VTP Domain Name : rsatechforum.in
VTP Pruning Mode : Disabled

VTP Traps Generation : Disabled
Device ID : aabb.cc80.0300

Feature VLAN:

VTP Operating Mode : Client

Number of existing VLANs : 5

Number of existing extended VLANs : 0

Maximum VLANs supported locally : 4096

Configuration Revision : 1

Primary ID : aabb.cc80.0100

Primary Description : SW-1

MD5 digest : 0x9A 0xCD 0x85 0xFC 0xB0 0xAA 0x1B 0x2D
0xB0 0x93 0x8E 0x49 0xD2 0x3B 0x63 0x8A

Feature MST:

VTP Operating Mode : Transparent

Feature UNKNOWN:

VTP Operating Mode : Transparent

Task-5: Create VLAN-10 as Sale, VLAN-20 as Production and VLAN-30 as Marketing on SW-1

```
SW-1#config t
SW-1(config)#vlan 10
SW-1(config-vlan)#name Sale
SW-1(config-vlan)#exit
SW-1(config)#vlan 20
SW-1(config-vlan)#name Production
SW-1(config-vlan)#exit
SW-1(config)#vlan 30
SW-1(config-vlan)#name Marketing
SW-1(config-vlan)#exit
SW-1(config)#exit
SW-1#
```

✓ Verification & Testing:

SW-1#sh vlan brief

VLAN Name	Status	Ports
1 default	active	Et0/0, Et0/1, Et0/2, Et0/3 Et1/0, Et1/2, Et1/3, Et2/0 Et2/1, Et2/2, Et2/3, Et3/0 Et3/1, Et3/2, Et3/3
10 Sale	active	
20 Production	active	

```

30 Marketing          active
1002 fddi-default      act/unsup
1003 trcrf-default     act/unsup
1004 fddinet-default   act/unsup
1005 trbrf-default     act/unsup

```

SW-1#sh vtp status

```

VTP Version capable      : 1 to 3
VTP version running      : 3
VTP Domain Name          : rsatechforum.in
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                 : aabb.cc80.0100

```

Feature VLAN:

```

-----
VTP Operating Mode       : Primary Server
Number of existing VLANs : 8
Number of existing extended VLANs : 0
Maximum VLANs supported locally : 4096
Configuration Revision   : 2
Primary ID                : aabb.cc80.0100
Primary Description       : SW-1
MD5 digest                : 0x02 0xDF 0x4F 0x3A 0xD1 0x21 0x0E 0x67
                          0x29 0x74 0x10 0x24 0x33 0x0A 0x31 0xC4

```

Feature MST:

```

-----
VTP Operating Mode       : Transparent

```

Feature UNKNOWN:

```

-----
VTP Operating Mode       : Transparent

```

SW-2#sh vlan brief

VLAN Name	Status	Ports
1 default	active	Et0/0, Et0/1, Et0/2, Et0/3 Et1/2, Et1/3, Et2/0, Et2/1 Et2/2, Et2/3, Et3/0, Et3/1 Et3/2, Et3/3

```

10 Sale          active
20 Production    active
30 Marketing      active
1002 fddi-default act/unsup
1003 trcrf-default act/unsup
1004 fddinet-default act/unsup
1005 trbrf-default act/unsup

```

SW-2#sh vtp status

```

VTP Version capable      : 1 to 3

```

VTP version running : 3
 VTP Domain Name : rsatechforum.in
 VTP Pruning Mode : Disabled
 VTP Traps Generation : Disabled
 Device ID : aabb.cc80.0200

Feature VLAN:

VTP Operating Mode : Client
 Number of existing VLANs : 8
 Number of existing extended VLANs : 0
 Maximum VLANs supported locally : 4096
 Configuration Revision : 2
 Primary ID : aabb.cc80.0100
 Primary Description : SW-1
 MD5 digest : 0x02 0xDF 0x4F 0x3A 0xD1 0x21 0x0E 0x67
 0x29 0x74 0x10 0x24 0x33 0x0A 0x31 0xC4

Feature MST:

VTP Operating Mode : Transparent

Feature UNKNOWN:

VTP Operating Mode : Transparent

SW-3#sh vlan brief

VLAN Name	Status	Ports
1 default	active	Et0/0, Et0/1, Et0/2, Et0/3 Et1/1, Et1/2, Et1/3, Et2/0 Et2/1, Et2/2, Et2/3, Et3/0 Et3/1, Et3/2, Et3/3
10 Sale	active	
20 Production	active	
30 Marketing	active	
1002 fddi-default	act/unsup	
1003 trcrf-default	act/unsup	
1004 fddinet-default	act/unsup	
1005 trbrf-default	act/unsup	

SW-3#sh vtp status

VTP Version capable : 1 to 3
 VTP version running : 3
 VTP Domain Name : rsatechforum.in
 VTP Pruning Mode : Disabled
 VTP Traps Generation : Disabled
 Device ID : aabb.cc80.0300

Feature VLAN:

VTP Operating Mode : Client
Number of existing VLANs : 8
Number of existing extended VLANs : 0
Maximum VLANs supported locally : 4096
Configuration Revision : 2
Primary ID : aabb.cc80.0100
Primary Description : SW-1
MD5 digest : 0x02 0xDF 0x4F 0x3A 0xD1 0x21 0x0E 0x67
0x29 0x74 0x10 0x24 0x33 0x0A 0x31 0xC4

Feature MST:

VTP Operating Mode : Transparent

Feature UNKNOWN:

VTP Operating Mode : Transparent

Task-6: Assign switchports in respective VLAN as per the topology

Switch-1 configuration:

```
SW-1#config t
SW-1(config)#int e0/1
SW-1(config-if)#switchport mode access
SW-1(config-if)#switchport access vlan 10
SW-1(config-if)#exit
SW-1(config)#int e0/2
SW-1(config-if)#switchport mode access
SW-1(config-if)#switchport access vlan 20
SW-1(config-if)#exit
SW-1(config)#int e0/3
SW-1(config-if)#switchport mode access
SW-1(config-if)#switchport access vlan 30
SW-1(config-if)#exit
SW-1(config)#exit
SW-1#
```

Switch-2 configuration:

```
SW-2#config t
SW-2(config)#int e0/0
SW-2(config-if)#switchport mode access
SW-2(config-if)#switchport access vlan 10
SW-2(config-if)#exit
SW-2(config)#int e0/1
SW-2(config-if)#switchport mode access
SW-2(config-if)#switchport access vlan 20
SW-2(config-if)#exit
SW-2(config)#int e0/2
```

```
SW-2(config-if)#switchport mode access
SW-2(config-if)#switchport access vlan 30
SW-2(config-if)#exit
SW-2(config)#exit
SW-2#
```

Switch-3 configuration:

```
SW-3#config t
SW-3(config)#int e0/0
SW-3(config-if)#switchport mode access
SW-3(config-if)#switchport access vlan 10
SW-3(config-if)#exit
SW-3(config)#int e0/1
SW-3(config-if)#switchport mode access
SW-3(config-if)#switchport access vlan 20
SW-3(config-if)#exit
SW-3(config)#int e0/2
SW-3(config-if)#switchport mode access
SW-3(config-if)#switchport access vlan 30
SW-3(config-if)#exit
SW-3(config)#exit
SW-3#
```

✓ Verification & Testing:

SW-1#sh vlan brief

VLAN Name	Status	Ports
1 default	active	Et0/0, Et1/0, Et1/2, Et1/3 Et2/0, Et2/1, Et2/2, Et2/3 Et3/0, Et3/1, Et3/2, Et3/3
10 Sale	active	Et0/1
20 Production	active	Et0/2
30 Marketing	active	Et0/3
1002 fddi-default	act/unsup	
1003 trcrf-default	act/unsup	
1004 fddinet-default	act/unsup	
1005 trbrf-default	act/unsup	

SW-1#sh vlan summary

```
Number of existing VLANs : 8
Number of existing VTP VLANs : 8
Number of existing extended VTP VLANs : 0
```

SW-2#sh vlan brief

VLAN Name	Status	Ports
1 default	active	Et0/3, Et1/2, Et1/3, Et2/0

Et2/1, Et2/2, Et2/3, Et3/0
Et3/1, Et3/2, Et3/3

10	Sale	active	Et0/0
20	Production	active	Et0/1
30	Marketing	active	Et0/2
1002	fddi-default	act/unsup	
1003	trcrf-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trbrf-default	act/unsup	

SW-2#sh vlan summary

Number of existing VLANs : 8
Number of existing VTP VLANs : 8
Number of existing extended VTP VLANs : 0

SW-3#sh vlan brief

VLAN Name	Status	Ports
1 default	active	Et0/3, Et1/1, Et1/2, Et1/3 Et2/0, Et2/1, Et2/2, Et2/3 Et3/0, Et3/1, Et3/2, Et3/3
10 Sale	active	Et0/0
20 Production	active	Et0/1
30 Marketing	active	Et0/2
1002 fddi-default	act/unsup	
1003 trcrf-default	act/unsup	
1004 fddinet-default	act/unsup	
1005 trbrf-default	act/unsup	

SW-3#sh vlan summary

Number of existing VLANs : 8
Number of existing VTP VLANs : 8
Number of existing extended VTP VLANs : 0

Task-7: Check connectivity between PCs in same VLAN**Ping PC-2 and PC-3 from PC-1****PC1> ping 192.168.1.11**

84 bytes from 192.168.1.11 icmp_seq=1 ttl=64 time=3.434 ms
84 bytes from 192.168.1.11 icmp_seq=2 ttl=64 time=3.230 ms
84 bytes from 192.168.1.11 icmp_seq=3 ttl=64 time=1.998 ms
84 bytes from 192.168.1.11 icmp_seq=4 ttl=64 time=3.646 ms
84 bytes from 192.168.1.11 icmp_seq=5 ttl=64 time=3.510 ms

PC1> ping 192.168.1.12

84 bytes from 192.168.1.12 icmp_seq=1 ttl=64 time=1.675 ms
84 bytes from 192.168.1.12 icmp_seq=2 ttl=64 time=5.777 ms
84 bytes from 192.168.1.12 icmp_seq=3 ttl=64 time=5.776 ms
84 bytes from 192.168.1.12 icmp_seq=4 ttl=64 time=7.706 ms

84 bytes from 192.168.1.12 icmp_seq=5 ttl=64 time=7.451 ms

Ping PC-5 and PC-6 from PC-4

PC4> ping 192.168.2.11

84 bytes from 192.168.2.11 icmp_seq=1 ttl=64 time=5.861 ms
 84 bytes from 192.168.2.11 icmp_seq=2 ttl=64 time=5.239 ms
 84 bytes from 192.168.2.11 icmp_seq=3 ttl=64 time=5.036 ms
 84 bytes from 192.168.2.11 icmp_seq=4 ttl=64 time=4.887 ms
 84 bytes from 192.168.2.11 icmp_seq=5 ttl=64 time=3.416 ms

PC4> ping 192.168.2.12

84 bytes from 192.168.2.12 icmp_seq=1 ttl=64 time=1.998 ms
 84 bytes from 192.168.2.12 icmp_seq=2 ttl=64 time=2.296 ms
 84 bytes from 192.168.2.12 icmp_seq=3 ttl=64 time=2.330 ms
 84 bytes from 192.168.2.12 icmp_seq=4 ttl=64 time=2.558 ms
 84 bytes from 192.168.2.12 icmp_seq=5 ttl=64 time=2.648 ms

Ping PC-8 and PC-9 from PC-7

PC7> ping 192.168.3.11

84 bytes from 192.168.3.11 icmp_seq=1 ttl=64 time=1.655 ms
 84 bytes from 192.168.3.11 icmp_seq=2 ttl=64 time=3.252 ms
 84 bytes from 192.168.3.11 icmp_seq=3 ttl=64 time=5.518 ms
 84 bytes from 192.168.3.11 icmp_seq=4 ttl=64 time=5.655 ms
 84 bytes from 192.168.3.11 icmp_seq=5 ttl=64 time=4.656 ms

PC7> ping 192.168.3.12

84 bytes from 192.168.3.12 icmp_seq=1 ttl=64 time=6.224 ms
 84 bytes from 192.168.3.12 icmp_seq=2 ttl=64 time=5.378 ms
 84 bytes from 192.168.3.12 icmp_seq=3 ttl=64 time=4.879 ms
 84 bytes from 192.168.3.12 icmp_seq=4 ttl=64 time=4.603 ms
 84 bytes from 192.168.3.12 icmp_seq=5 ttl=64 time=3.906 ms

Ping and telnet the SW-1 from Mgmt PC

```
C:\Users\Ratan>ping 150.150.150.1

Pinging 150.150.150.1 with 32 bytes of data:
Reply from 150.150.150.1: bytes=32 time<1ms TTL=255
Reply from 150.150.150.1: bytes=32 time=1ms TTL=255
Reply from 150.150.150.1: bytes=32 time=2ms TTL=255
Reply from 150.150.150.1: bytes=32 time<1ms TTL=255

Ping statistics for 150.150.150.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

Telnet to 150.150.150.1

```
Command Prompt

#####
##### Welcome to RSATechforum #####
##### This is Switch-1 #####
#####
#####

User Access Verification

Password:
SW-1>en
Password:
SW-1#

Connection to host lost.

C:\Users\Ratan>
```

Ping and telnet the SW-2 from Mgmt. PC

```
C:\Users\Ratan>ping 150.150.150.2

Pinging 150.150.150.2 with 32 bytes of data:
Reply from 150.150.150.2: bytes=32 time=1ms TTL=255
Reply from 150.150.150.2: bytes=32 time=3ms TTL=255
Reply from 150.150.150.2: bytes=32 time=2ms TTL=255
Reply from 150.150.150.2: bytes=32 time=2ms TTL=255

Ping statistics for 150.150.150.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 2ms
```

Telnet to 150.150.150.2

```
Telnet 150.150.150.2

#####
##### Welcome to RSATechforum #####
##### This is Switch-2 #####
#####
#####

User Access Verification

Password:
SW-2>en
Password:
SW-2#
```

Ping and telnet the SW-2 from Mgmt. PC

```
C:\Users\Ratan>ping 150.150.150.3

Pinging 150.150.150.3 with 32 bytes of data:
Reply from 150.150.150.3: bytes=32 time=1ms TTL=255
Reply from 150.150.150.3: bytes=32 time=1ms TTL=255
Reply from 150.150.150.3: bytes=32 time=1ms TTL=255
Reply from 150.150.150.3: bytes=32 time=2ms TTL=255

Ping statistics for 150.150.150.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms
```

Telnet to 150.150.150.3

```
CA: Telnet 150.150.150.3

#####
##### Welcome to RSATechforum #####
##### This is Switch-3 #####
#####

User Access Verification

Password:
SW-3>en
Password:
SW-3#
```

To test VTP feature, create VLAN-40 in SW-2 and SW-3 (Client mode)

```
SW-2#config t
SW-2(config)#vlan 40
```

VTP VLAN configuration not allowed when device is in CLIENT mode.

```
SW-2(config)#
```

```
SW-3#config t
SW-3(config)#vlan 40
```

VTP VLAN configuration not allowed when device is in CLIENT mode.

```
SW-3(config)#
```

Task-8: Configure SW-2 as VTP Transparent mode.

```
SW-2#config t
SW-2(config)#vtp mode transparent
```

Setting device to VTP Transparent mode for VLANS.

```
SW-2(config)#vtp domain rsatechforum.in
Domain name already set to rsatechforum.in.
```

```
SW-2(config)#vtp version 3
```

VTP version is already in V3.

```
SW-2(config)#vtp password Pass@123
```

Password already set to Pass@123

```
SW-2(config)#exit
```

```
SW-2#
```

Test: Create additional VLAN-40 on SW-1 (VTP-Server) and VLAN-50 on SW-2 (VTP Transparent) and observe the VTP information propagation on SW-2 and SW-3.

SW-1 Configuration:

```
SW-1#config t
SW-1(config)#vlan 40
SW-1(config-vlan)#name IT
SW-1(config-vlan)#exit
SW-1(config)#exit
SW-1#
```

SW-2 Configuration:

```
SW-2#config t
SW-2(config)#vlan 50
SW-2(config-vlan)#name Employee
SW-2(config-vlan)#exit
SW-2(config)#exit
SW-2#
```

✓ Verification & Testing:

SW-1#sh vlan brief

VLAN Name	Status	Ports
1 default	active	Et0/0, Et1/0, Et1/2, Et1/3 Et2/0, Et2/1, Et2/2, Et2/3 Et3/0, Et3/1, Et3/2, Et3/3
10 Sale	active	Et0/1
20 Production	active	Et0/2
30 Marketing	active	Et0/3
40 IT	active	
1002 fddi-default	act/unsup	
1003 trcrf-default	act/unsup	
1004 fddinet-default	act/unsup	
1005 trbrf-default	act/unsup	

```
SW-1#sh vtp status
VTP Version capable      : 1 to 3
VTP version running      : 3
VTP Domain Name          : rsatechforum.in
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                : aabb.cc80.0100
```

Feature VLAN:

```
-----
VTP Operating Mode       : Primary Server
Number of existing VLANs : 9
```

Number of existing extended VLANs : 0

Maximum VLANs supported locally : 4096

Configuration Revision : 3

Primary ID : aabb.cc80.0100

Primary Description : SW-1

MD5 digest : 0x4A 0x3D 0x40 0xFA 0xA0 0xF6 0xE4 0xBF
0xA8 0xDD 0xD4 0xC3 0x3E 0x5B 0xE2 0x9A

Feature MST:

VTP Operating Mode : Transparent

Feature UNKNOWN:

VTP Operating Mode : Transparent

SW-3#sh vlan brief

VLAN Name	Status	Ports
1 default	active	Et0/3, Et1/1, Et1/2, Et1/3 Et2/0, Et2/1, Et2/2, Et2/3 Et3/0, Et3/1, Et3/2, Et3/3
10 Sale	active	Et0/0
20 Production	active	Et0/1
30 Marketing	active	Et0/2
40 IT	active	
1002 fddi-default	act/unsup	
1003 trcrf-default	act/unsup	
1004 fddinet-default	act/unsup	
1005 trbrf-default	act/unsup	

SW-3#sh vtp status

VTP Version capable : 1 to 3

VTP version running : 3

VTP Domain Name : rsatechforum.in

VTP Pruning Mode : Disabled

VTP Traps Generation : Disabled

Device ID : aabb.cc80.0300

Feature VLAN:

VTP Operating Mode : Client

Number of existing VLANs : 9

Number of existing extended VLANs : 0

Maximum VLANs supported locally : 4096

Configuration Revision : 3

Primary ID : aabb.cc80.0100

Primary Description : SW-1

MD5 digest : 0x4A 0x3D 0x40 0xFA 0xA0 0xF6 0xE4 0xBF
0xA8 0xDD 0xD4 0xC3 0x3E 0x5B 0xE2 0x9A

Feature MST:

VTP Operating Mode : Transparent

Feature UNKNOWN:

VTP Operating Mode : Transparent

SW-2#sh vlan brief

VLAN Name	Status	Ports
1 default	active	Et0/3, Et1/2, Et1/3, Et2/0 Et2/1, Et2/2, Et2/3, Et3/0 Et3/1, Et3/2, Et3/3
10 Sale	active	Et0/0
20 Production	active	Et0/1
30 Marketing	active	Et0/2
50 Employee	active	
1002 fddi-default	act/unsup	
1003 trcrf-default	act/unsup	
1004 fddinet-default	act/unsup	
1005 trbrf-default	act/unsup	

SW-2#sh vtp status

VTP Version capable : 1 to 3
VTP version running : 3
VTP Domain Name : rsatechforum.in
VTP Pruning Mode : Disabled
VTP Traps Generation : Disabled
Device ID : aabb.cc80.0200

Feature VLAN:

VTP Operating Mode : Transparent

Number of existing VLANs : 9
Number of existing extended VLANs : 0
Maximum VLANs supported locally : 4096

Feature MST:

VTP Operating Mode : Transparent

Feature UNKNOWN:

VTP Operating Mode : Transparent

Important Commands:

Sh vlan brief
Sh vlan summary
Sh vtp status

Ratan